

CHAPTER I

INTRODUCTION AND DESIGN OF THE STUDY

1.1 INTRODUCTION OF THE STUDY

Internships form an essential part of postgraduate academic learning, bridging the gap between theoretical knowledge and practical experience. As part of the M.Com (FCA) curriculum at Bharathiar University, a 45-day internship was undertaken at Podaran Foods India Private Limited, a prominent food manufacturing company in Tamil Nadu. This internship offered first hand exposure to the day-to-day functions and operations of key departments such as Production, Manufacturing and Packaging departments in the company.

Podaran Foods India Private Limited specializes in the production of soft drinks and packaged foods. As an observer intern, the study focused on understanding the workflows, safety standards, quality control practices, and operational efficiency across the three major departments. The experience allowed observation of modern industrial practices, labour management, supply chain processes, and compliance with food safety regulations, all of which are integral to managing and understanding manufacturing operations in the FMCG sector. This report documents the learning outcomes, departmental analysis, challenges observed, and the practical applications relevant to commerce and industrial operations.

Podaran Foods primarily specializes in the manufacturing of carbonated soft drinks, fruit-based beverages, flavoured drinks, milk-based beverages, and packaged drinking water. Some of the popular carbonated products manufactured include Paneer Soda, Lemon Soda, Cola, Orange Soda, and Power Cola. In the fruit beverage segment, the company produces Mango, Guava, Apple, and Litchi juices, which are either pulp-based or essence-based depending on the formulation. Under the flavoured beverage category, Podaran manufactures drinks with synthetic flavours like rose, grape, and jeera. The milk-based products include flavoured milk varieties such as Chocolate Milk and Strawberry Milk, which are especially popular among children. The company also produces limited packaged snacks such as salted groundnuts and chips, which are distributed along with its beverages in select markets.

The learning experience covered multiple aspects of factory operations, including production planning, machinery handling, bottle manufacturing (blowing), filling, labelling, quality control, and dispatch coordination. Observing these departments helped me understand the importance of teamwork, process efficiency, cleanliness, and technical supervision in a food-grade manufacturing environment. This report is based on real-time exposure to the workflow of Podaran Foods and aims to document my observations, learnings, and professional growth during the internship.

1.2 AREA OF TRAINING

I have trained in the following department

- Production department
- Manufacturing department
- Quality control department
- Packaging department

1.3 PERIOD OF STUDY

I have undergone internship training at Podaran Foods India Private limited for a period of 45 days from 09/05/2025 to 30/06/2025.

CHAPTER II

COMPANY PROFILE

2.1 INTRODUCTION OF THE COMPANY

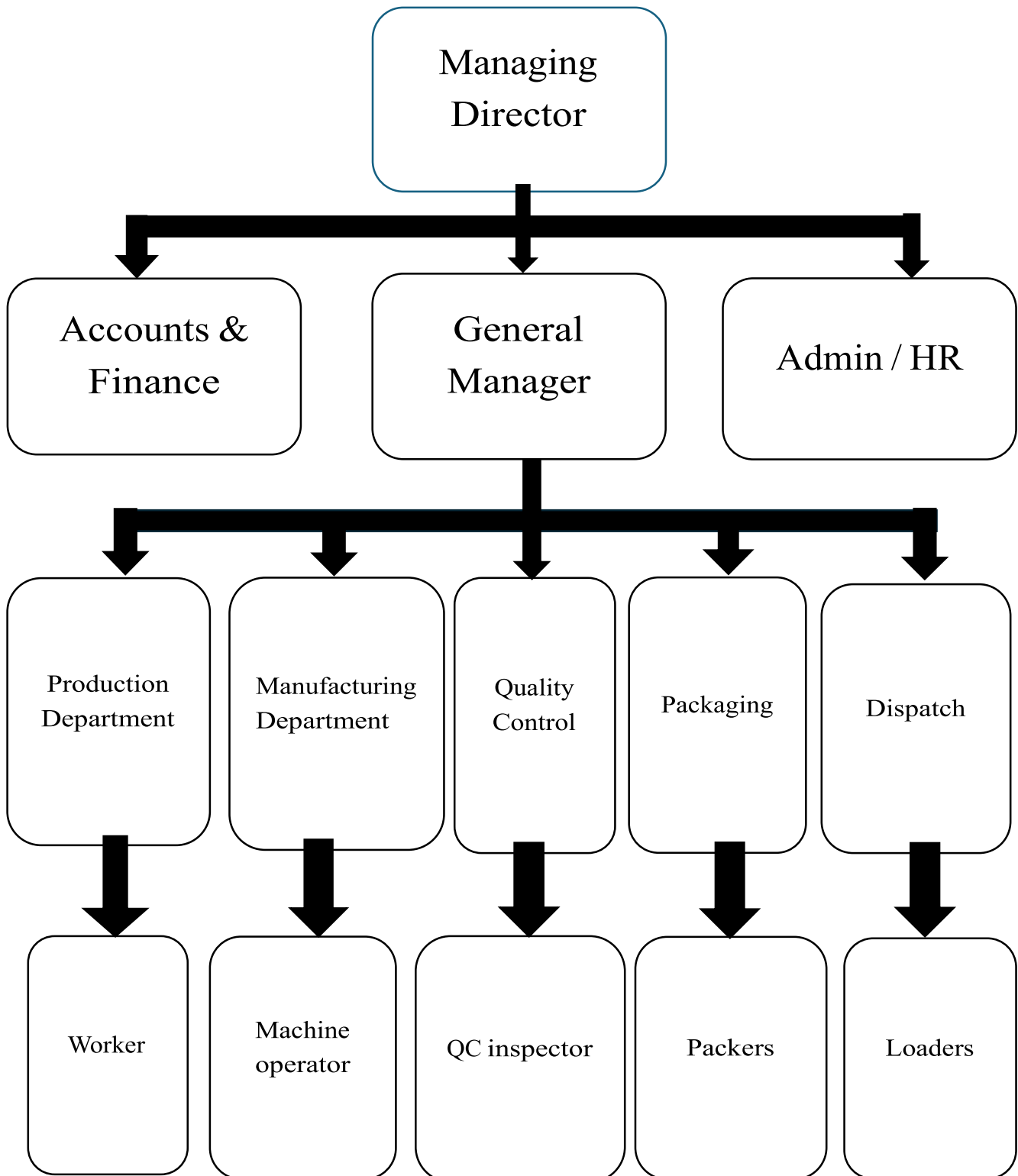
Podaran Foods India Private Limited is a Tamil Nadu-based private limited company engaged in the manufacture of fruit-based beverages, specializing in the production of hygienically processed and packaged fruit juices. Located in Melapalayam, Nakakadayur, the company has earned a reputation in the regional market for its quality-focused production and cost-effective distribution of natural juice products. The company was founded with a clear mission to bridge the gap between traditional fruit consumption and modern beverage convenience. In line with this, Podaran Foods has designed its operations around a semi-automated production line that emphasizes efficiency, hygiene, and consistency. The factory is equipped with specialized equipment for fruit washing, pulp extraction, filtration, pasteurization, blending, bottling, sealing, labelling, and final packaging. This systematic process ensures that each batch meets the company's internal quality benchmarks and complies with FSSAI (Food Safety and Standards Authority of India) regulations.

Podaran Foods also emphasizes worker safety, environmental cleanliness, and resource efficiency, aligning itself with responsible manufacturing practices. The factory operates on a twoshift schedule, enabling continuous production while ensuring effective supervision and maintenance. Podaran Foods is guided by the core values of integrity, innovation, customer satisfaction, and quality assurance. The management fosters a collaborative work environment, where cross-functional coordination between production, packaging, and logistics teams ensures timely and safe delivery of finished goods. By continually investing in its people, processes, and product development, Podaran Foods India Private Limited aspires to become a leading player in the Indian juice market. It remains committed to contributing to public health, economic development, and responsible manufacturing through every bottle it produces. By combining local sourcing of raw materials, a dedicated team, and a strong commitment to food safety, Podaran Foods India Private Limited has positioned itself as a promising player in South India's juice manufacturing sector. The organization continues to grow steadily by expanding its product offerings and strengthening its presence in both urban and rural retail markets.

2.2 COMPANY PROFILE

PARTICULARS	DETAILS
Name of the company	Podaran Foods India Private Limited
Address	461/1, Melapalayam, Chitampalayam, Nathakadayur via, Kangayam Taluk, Tirupur District, Tamil Nadu – 638108
Head office	Melapalayam, Nakakadayur, Tirupur District, Tamil Nadu
Managing director	Mr. P. Suresh Kumar
Business type	Private Limited Company
Website	https://www.podaranfoods.com/
Accredited	FSSAI Approved, ISO 22000 (Food Safety)
Industry	Food & Beverage – Fruit Juices & Soft Drink Manufacturing
GST no.	33AAHCP3183F1ZJ
Establishment year	1969
CIN no.	U15400TZ2013PTC019718
Email	md@podaranfoods.com
Logo	

2.3 ORGANIZATIONAL CHART



2.4 VISION AND MISSION

VISION

“To become a leading food and beverage manufacturer in South India, offering safe, affordable, and hygienically processed products that satisfy customer needs and taste preferences.”

MISSION

- To consistently improve food safety and manufacturing efficiency.
- To invest in semi-automated technology and trained workforce.
- To deliver products that meet both local taste and quality expectations.
- To follow environmentally conscious and sustainable practices in operations.

2.5 CORE VALUES

- **Quality:** The company is committed to maintaining high quality standards through regular testing and strict adherence to food safety protocols. Every product is checked for taste, hygiene, and consistency.
- **Customer Satisfaction:** Customer needs and preferences are prioritized. Products are made affordable and tailored to regional tastes, with a focus on timely delivery and appealing packaging.
- **Hygiene and Safety:** Strict hygiene practices are followed in all departments. Employees use protective gear and production areas are cleaned regularly to ensure safe food handling.
- **Teamwork and Collaboration:** Podaran believes that successful production and delivery of goods require effective teamwork. Departments such as Production, Manufacturing, Quality Control, Packaging, and Logistics work together in coordination.
- **Innovation:** While Podaran retains its traditional values and taste legacy, it also embraces innovation. The company regularly upgrades its machinery, packaging systems, and product offerings to remain competitive. During the internship, I noticed trials being run for new flavors and improvements in shrink-wrapping equipment. This shows the company’s adaptability to market demands and technological advancements.

2.6 PRODUCTS OF THE COMPANY

2.6.1 CARBONATED SOFT DRINKS



Carbonated soft drinks are one of the major products manufactured by Podaran Foods India Pvt. Ltd. These beverages are made by blending purified water with flavoured syrup and carbon dioxide gas (CO₂) under controlled pressure. The resulting drink is fizzy, refreshing, and available in various regional flavours to suit consumer preferences.

Flavours Produced:

- Cola
- Orange
- Lemon
- Mango
- Paneer soda
- Jeera masala
- Power cola

2.6.2. FRUIT JUICES



Podaran Foods India Pvt. Ltd. Manufactures a variety of non-carbonated fruit juices that are made using real fruit pulp or concentrates. These juices are sweetened, pasteurized, and packed under hygienic conditions to ensure taste, safety, and shelf life. They are especially popular among health-conscious consumers, schoolchildren, and families. The juices are available in different flavors and are filled into PET bottles ranging from 200 ml to 1 liter.

Types of Fruit Juices Produced:

1. Mango
2. Apple
3. Guava
4. Mixed fruit

2.6.3 FLAVOURED DRINKS



Flavoured drinks are a popular product segment manufactured by Podaran Foods India Pvt. Ltd. These beverages are especially appealing to regional markets that enjoy traditional and health-oriented soft drinks. The company focuses on providing refreshing options that not only quench thirst but also offer functional benefits such as digestion and cooling.

The flavoured drinks are typically mildly carbonated, hygienically processed, and attractively packaged. These drinks are in high demand during summer seasons and festive occasions.

Types of flavoured drink produced:

1. Paneer soda
2. Jeera masala
3. Lemonade

2.6.4 MILK-BASED PRODUCTS



Podaran Foods India Pvt. Ltd. Also operates a dairy beverage division that produces a range of milk-based drinks. These beverages are rich in taste, nutritious, and cater to the growing demand for healthy, ready-to-drink dairy options. Milk-based beverages are produced in flavored variants, enriched with milk solids, sugar, and permitted natural or nature-identical flavors. They are especially consumed by children, students, and those looking for an energy drink during work or travel.

All milk drinks are prepared under pasteurized conditions and stored in chilled environments to maintain safety and quality.

Types of Milk-Based Beverages:

1. Badam milk
2. Pista milk
3. Chocolate milk
4. Strawberry milk

CHAPTER III

DEPARTMENTS OF THE COMPANY

3.1 DEPARTMENTS OF THE COMPANY

Podaran Foods India Pvt. Ltd., located in Nathakadaiyur, Tamil Nadu, is engaged in the manufacturing and distribution of beverages such as carbonated soft drinks, fruit juices, flavoured water, and milk-based drinks. The company operates through a structured departmental workflow to ensure smooth and hygienic operations from raw material intake to final dispatch.

During my internship, I had the opportunity to focus on the core operational departments involved in real-time beverage processing. These departments are interlinked and work systematically to convert raw materials into packaged, ready-to-sell beverages.

3.2 DEPARTMENTS PRESENT IN THE COMPANY

The company has the following departments functioning under one integrated manufacturing facility:

1. Raw Material Storage & Inventory
2. Production Department
3. Manufacturing Department
4. Quality Control Department
5. Packaging Department
6. Dispatch & Logistics Department
7. Maintenance Department
8. Procurement / Purchase Department
9. Accounts & Billing Department
10. Administration / HR Department

3.3 DEPARTMENTS I FOCUSED ON DURING INTERNSHIP

I was guided through and observed the following departments in detail:

- Production Department
- Manufacturing Department
- Quality Control Department
- Packaging department

These departments form the backbone of Podaran's beverage processing unit and are crucial for ensuring quality, hygiene, and efficiency in day-to-day operations.

3.3.1 DEPARTMENTS I FOCUSED ON DURING INTERNSHIP

The Production Department Is where the journey of a Podaran drink begins. It involves the preparation of raw materials such as sugar, fruit pulp, RO water, flavouring essence, and preservatives. These are then mixed to form the base juice or soft drink syrup. The department plays a critical role as it determines the taste, consistency, and stability of the beverage before it goes into bottling The Production Department Is where the journey of a Podaran drink begins. It involves the preparation of raw materials such as sugar, fruit pulp, RO water, flavouring essence, and preservatives. These are then mixed to form the base juice or soft drink syrup. The department plays a critical role as it determines the taste, consistency, and stability of the beverage before it goes into bottling. (e.g., Paneer Soda, Cola, Lemon, etc.). For thick juices like Guava or Mango, real pulp is used, which is procured from suppliers in Bangalore and other regions.

At Podaran, the production area is neat, temperature-controlled, and organized, with designated storage areas for raw materials. The production team follows SOPs (Standard Operating Procedures) and works with a combination of manual handling and semi-automated mixing systems. Hygiene is strictly maintained here. All workers wear gloves, head caps, and coats. The area is cleaned regularly, and standard operating procedures (SOPs) are followed for every batch.

Objectives of the Production Department:

- To prepare safe and consistent beverage syrup
- To maintain hygiene during the raw material mixing process
- To ensure accurate ingredient measurement and filtration
- To store prepared syrup in clean, temperature-controlled tanks



STORAGE ROOM

Daily Process Flow:

1.Receipt of Raw Materials

Raw materials like fruit pulp (mango, guava, etc.), sugar sacks, essence bottles, and citric acid arrive and are checked for quality.

2. Storage & Sorting

Materials are stored in a clean warehouse. Labels and usage instructions are marked. Workers wear caps and gloves during handling.

3.Mixing Process

A large stainless-steel mixing tank (over 12 feet high) is used. Workers climb built-in stairs to pour in sugar and essence are marked. Workers wear caps and gloves during handling.

4.Batch Preparation Time

Each product takes about 10–12 hours to prepare and is usually made for 2 continuous days per batch. For example, if Paneer Soda is prepared today and tomorrow, Cola or Guava might be prepared the next day.

5. Filtration

The mixed juice passes through an industrial filtration unit to remove impurities and ensure it meets clarity standards.

6.Cooling & Transfer

The filtered juice is transferred to chilled storage tanks that maintain stable temperatures and prevent early spoilage.



MIXING PROCESS

Machinery & Equipment Used :

- Stainless Steel Syrup Mixers (12+ feet tall)
- Manual weighing scale for sugar/essence

- Filtration System with micron filters
- Cooled Storage Tanks (pressurized, insulated)

Safety, Hygiene & Team Practices

All staff in the production unit wear:

- Head caps
- Gloves
- Apron-style coats (These prevents contamination)
- The production area is mopped regularly.

3.3.2 MANUFACTURING DEPARTMENT



MOULDING

Once the beverage syrup is prepared in the production department, it moves to the Manufacturing Department, which is responsible for transforming the syrup into a finished product. This includes bottle blowing, cooling, juice filling, and initial conveyor checks.

Podaran Foods has semi-automated systems in this unit, and it was one of the most exciting departments I visited because I saw bottles being “born” from tiny preforms, then cooled and filled with fresh juice. The manufacturing department is fast-paced and focused on precision, hygiene, and consistency.

Objectives of the Manufacturing Department

- To blow PET preforms into bottles of required sizes (typically 200 ml)
- To cool down the newly formed bottles before filling
- To fill each bottle with the correct beverage according to batch schedule
- To monitor machines for faults, fill-level issues, or delays
- To maintain hygienic practices throughout the production cycle



BOTTLE MANUFACTURING

Process Flow I Observed:

1. Bottle Blowing Process:

I saw thousands of tiny, thick plastic pieces (called preforms) about the size of a pinky finger. These are placed into a blowing machine which heats the preforms and stretches them into 200 ml PET bottles using compressed air and moulds.

2. Bottle Cooling System:

Newly blown bottles are hot and must be cooled before filling. They are sent through a cooling tunnel, and I saw this through a glass panel. If bottles are not cooled properly, the juice's flavour and shelf life can be affected.

3. Juice Filling Process:

After cooling, bottles are moved by conveyor belts to the filling area. Here, juice from the chilled tanks is filled into the bottles via automated filling heads.

4. Belt Monitoring & Staff Duties:

Employees are stationed near the machines to check for any bottle jams, spills, or errors. If a machine stalls or malfunctions (like a stuck bottle), a beeping alert sound. I saw this happen once and a staff member quickly fixed it.

Machines & Technology Used:

- Blow Moulding Machine (for PET bottle formation)
- Compressed Air Compressor (for bottle expansion)
- Bottle Cooling Tunnel
- Juice Filling Line (semi-automatic, high-speed)
- Conveyor Belt Monitoring Sensors

Safety, Cleanliness & Standards:

- The area is dry-mopped frequently to avoid spills
- Workers use non-slip shoes, gloves, and head caps
- Machines are cleaned after each juice batch, especially nozzles and belts
- Temperature and bottle GSM are monitored regularly

Personal Observations:

- I was amazed to see how tiny preforms turn into bottles—it was like watching a real-time transformation.
- The bottle cooling was essential, and they ensured no heat entered the filling phase.
- Each filling head worked precisely, and the staff reacted quickly to any issues.
- I learned that even a single small jam in the conveyor can stop the entire flow.

What I Learned from the Manufacturing Department:

This department helped me understand how important precision and machine coordination are in beverage manufacturing. From blowing to cooling to filling, every second counts. Even a 0.2g difference in bottle thickness has meaning. I also realized how automation supports speed and quality while human supervision ensures safety and control.

3.3.3 QUALITY CONTROL DEPARTMENT



QUALITY CHECK

The Quality Control (QC) Department is one of the most crucial departments in any food and beverage manufacturing company — and Podaran Foods India Pvt. Ltd. is no exception. This department ensures that every drink manufactured is safe, hygienic, and consistent in taste, volume, and appearance before it reaches the consumer.

Although I wasn't taken into a laboratory, I observed multiple quality control measures integrated directly on the factory floor, including filtration checks, sensor-based volume validation, and manual inspections for cleanliness and accuracy. This shows that quality at Podaran is not just a step but a continuous process.

Objectives of the Quality Control Department:

- To ensure the final product is safe for human consumption
- To verify accurate volume and consistency of each bottle
- To check hygiene and sanitation during production
- To inspect bottles for physical defects or errors
- To validate that labelling and dates are accurate and legible

Processes I Observed Under Quality Control:

1. Filtration as First Quality Gate:

The first major quality control checkpoint happens in the Production Department itself — via a filtration unit. This system removes any undissolved particles, clumps, or contaminants that may remain after mixing.

2. Bottle Volume Verification:

After filling, each bottle is scanned by a sensor-based system that checks if it contains the correct amount — exactly 200 ml for standard-size products.

3. Manual Observations by Staff:

Near every critical point (filling, capping, labelling), workers continuously watch:

- Loose or missing caps
- Spills on the bottle body
- Discoloration or turbidity in the liquid
- Any signs of leakage

4. Label and Print Check:

At the labelling station, I noticed that the MRP, batch number, manufacturing date, and use-by date are printed using a high-speed machine. However, final legibility checks are done by both sensors and human workers to ensure compliance with labelling standards.

Machines and Tools Involved in Quality Control:

- Industrial Filtration Machine (micron level)
- Volume Sensor System

- Label Printing Gun (for date, MRP)
- Defect Rejection Arms (mechanical pushers)
- Manual Inspection Lights at QC stations

Personal Observations:

- I was surprised at how precise and automatic the QC process was.
- Even if a bottle had only a small defect or under filled by a few ml, it was still rejected.
- The labelling gun was fast, and the batch coding was printed clearly.
- Workers paid close attention and corrected problems quickly without interrupting the flow.

3.3.4 PACKAGING DEPARTMENT



LABELLING

The Packaging Department at Podaran Foods India Pvt. Ltd. Is where the final transformation of the product takes place — from a plain filled bottle to a branded, properly labelled and market-ready product. This department ensures that each beverage is not only

protected for storage and transport but also appeals to consumers through professional presentation.

Packaging is both a technical and visual process. It combines machinery (like shrink wrappers, printers, conveyors) and manual supervision to deliver accuracy, hygiene, and branding compliance.

Objectives of the Packaging Department:

- To label each bottle with correct branding and product details
- To print batch number, manufacturing date, expiry, and MRP
- To detect and remove defective or damaged bottles
- To group bottles into sets and wrap them for safe handling
- To prepare the finished product for warehouse storage and dispatch

Process Flow I Observed in Packaging:

1. Bottle Transfer from Filling to Packaging Line

After the bottles are filled and capped, they are transferred via conveyor belts into the packaging zone. I noticed that separate belts are used for different product types to avoid confusion or misplacement.

2. Labelling Machines in Action

Each bottle is passed through a high-speed labelling machine that applies branded labels (e.g., for Paneer Soda, Lemon Soda, Cola). These labels contain:

- Brand name and logo
- Ingredients list
- Product volume (e.g., 200 ml)
- Manufacturing company details

3. MRP and Date Printing

I saw a gun-like printing device that stamps:

- Manufacturing Date
- Expiry/Use by Date

- Maximum Retail Price (MRP). This happens automatically as bottles pass under the printer. Accuracy is checked both by sensor and manual visual check.

4. Defect Detection & Rejection

After labelling and printing, a quality sensor checks for:

- Improperly stuck labels
- Blurred printing
- Scratched or dented bottles

5. Shrink Wrapping into Sets

I saw that bottles are grouped into sets of 12 using a wrapping machine. A thin plastic film wraps these sets tightly and evenly, preparing them for safe stacking and transport.

6. Pallet Stacking and Warehouse Transfer

Wrapped sets are then arranged on pallets using a semi-automated stacker or manually by workers. Pallets are sorted based on product type and moved to the warehouse where they await loading into delivery trucks.



STACKING

Machines and Tools Involved:

- Conveyor Belt System
- Automatic Labelling Machine

- Label Gun
- Defect Sensor & Rejection Arm
- Shrink Wrapping Machine
- Pallet Stacker (semi-robotic)

Personal Observations:

- I was surprised at the speed and precision of the labelling and printing machines they were so fast that it was difficult to follow with my eyes!
- When one label was slightly bent, the sensor beeped, and the bottle was removed instantly.
- The wrapping machine grouped 12 bottles with tight, clean seals, and workers stacked them efficiently.
- Bottles were sorted based on flavour and clearly marked for transport.

What I Learned from the Packaging Department:

This department showed me that the final presentation of a product is as important as its internal quality. Even a tasty drink won't sell if it looks unclean, mislabelled, or damaged. Packaging adds value, safety, and consumer trust. I also understood how machinery and human monitoring must work side-by-side to ensure efficiency without sacrificing quality.

CHAPTER IV

KNOWLEDGE GAINED AND CONCLUSION

4.1 KNOWLEDGE GAINED

- I gained practical exposure to how beverages like soft drinks and juices are manufactured in an industrial setting.
- I observed how raw materials are measured, mixed, and processed to produce consistent and safe products.
- I understood how semi-automated machinery like blow moulding units, filling machines, and labelling systems operate in coordination.
- I learned the importance of maintaining hygiene standards and how quality control is embedded at every stage of production.
- I experienced how packaging is done using labelling machines, MRP/date printers, and shrink-wrapping equipment.
- I understood the role of teamwork and how different departments collaborate to achieve efficient production flow.
- I realized how theoretical concepts from commerce and industry studies are applied in real time factory environments.

4.2 CONCLUSION

My internship experience at Podaran Foods India Pvt. Ltd. Was an enriching journey. It gave me a complete overview of the beverage manufacturing process, including departments such as Production, Manufacturing, Quality Control, and Packaging. The well-coordinated workflow, attention to safety and hygiene, and the combination of machinery with human oversight helped me learn how modern industries operate. This hands-on exposure gave me clarity about the expectations and standards of working in a production environment. It also helped me understand the importance of discipline, process knowledge, hygiene and teamwork.